

MILESTONE 2 PROGRESS UPDATE

Macy's Marketing Campaign Execution

GenAI Co-worker for Omnichannel Retail Operations

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From Workflow to Skills to Data

10 STEP WORKFLOW

1. Briefing
2. Segmentation
3. SKU Selection
4. Creative Production
5. Layout Assembly
6. Approval
7. Localization
8. Activation
9. Monitoring
10. Reporting

4 SKILLS

SKILL 1: Audience Segment Builder

Targets: Step 2

SKILL 2: DAM Asset Finder

Targets: Steps 4 to 5

SKILL 3: Localization Generator

Targets: Step 7

SKILL 4: Campaign Performance Analyzer

Targets: Step 9

DATA NEEDED

- Customer profiles, transactions, loyalty data
- DAM asset metadata, tags, rights, resolution
- Master ad spec, regional pricing, inventory
- Campaign performance by channel and segment

Each skill solves one workflow bottleneck. Each one needs specific data.

The Database We Built

SQLite, ~480K rows across 7 tables, fully reproducible

7 TABLES

Table	Rows
customers	50,000
transactions	417,180
sku_catalog	2,000
regional_pricing	20,000
dam_assets	5,000
campaigns	12
campaign_performance	2,920

SAMPLE: customers TABLE (top spenders)

customer_id	name	age	loyalty_tier	total_spend
48891	Sarah Shaw	32	Platinum	\$12,357.71
49002	Veronica Williams	51	Platinum	\$11,238.15
40408	Philip Smith	71	Platinum	\$10,842.03
3403	Gabriel Jenkins	69	Platinum	\$10,625.94
21087	Jennifer Tran	70	Platinum	\$10,398.69

Real product photos pulled from Unsplash API.

Mess simulated on purpose: 2% null emails, expired DAM rights, mislabeled assets.

Seeded for reproducibility.

Database lives at `data/macys.db`. Every row above pulled live from the file.

How the Data Drives a Skill: Audience Segment Builder

SARAH'S QUESTION

“Who should see the Mother's Day Beauty campaign?”

THE SKILL QUERIES THE DATABASE

```
SELECT c.customer_id, c.age, c.loyalty_tier,  
       MAX(t.transaction_date) AS last_purchase_date  
FROM customers c  
JOIN transactions t ON c.customer_id = t.customer_id  
JOIN sku_catalog s ON t.sku_id = s.sku_id  
WHERE s.category = 'Beauty'  
       AND t.transaction_date > '2025-04-01'  
GROUP BY c.customer_id, c.age, c.loyalty_tier  
LIMIT 5;
```

REAL OUTPUT FROM OUR DATABASE

customer_id	age	loyalty_tier	last_purchase_date
1	18	Bronze	2025-09-18
2	37	Gold	2025-09-17
3	69	Silver	2025-11-04
5	56	Bronze	2025-09-09
6	66	Bronze	2025-12-29

4 hours of manual Excel work becomes 90 seconds. The skill speaks the same language Sarah speaks.

What is Next

BUILDING THIS WEEK

- Skill 1: Audience Segment Builder
- Skill 2: DAM Asset Finder
- Skill 3: Localization Generator
- Skill 4: Campaign Performance Analyzer
- README mapping every workflow task to its skill, data, and code

DEMO ON MAY 7

- 10-minute video walkthrough
- Live skill execution against the database
- Real outputs for the Mother's Day campaign scenario
- Team narration without scripts

Repository: github.com/rsm-genai-2026/mgt449-milestone02-Group-4